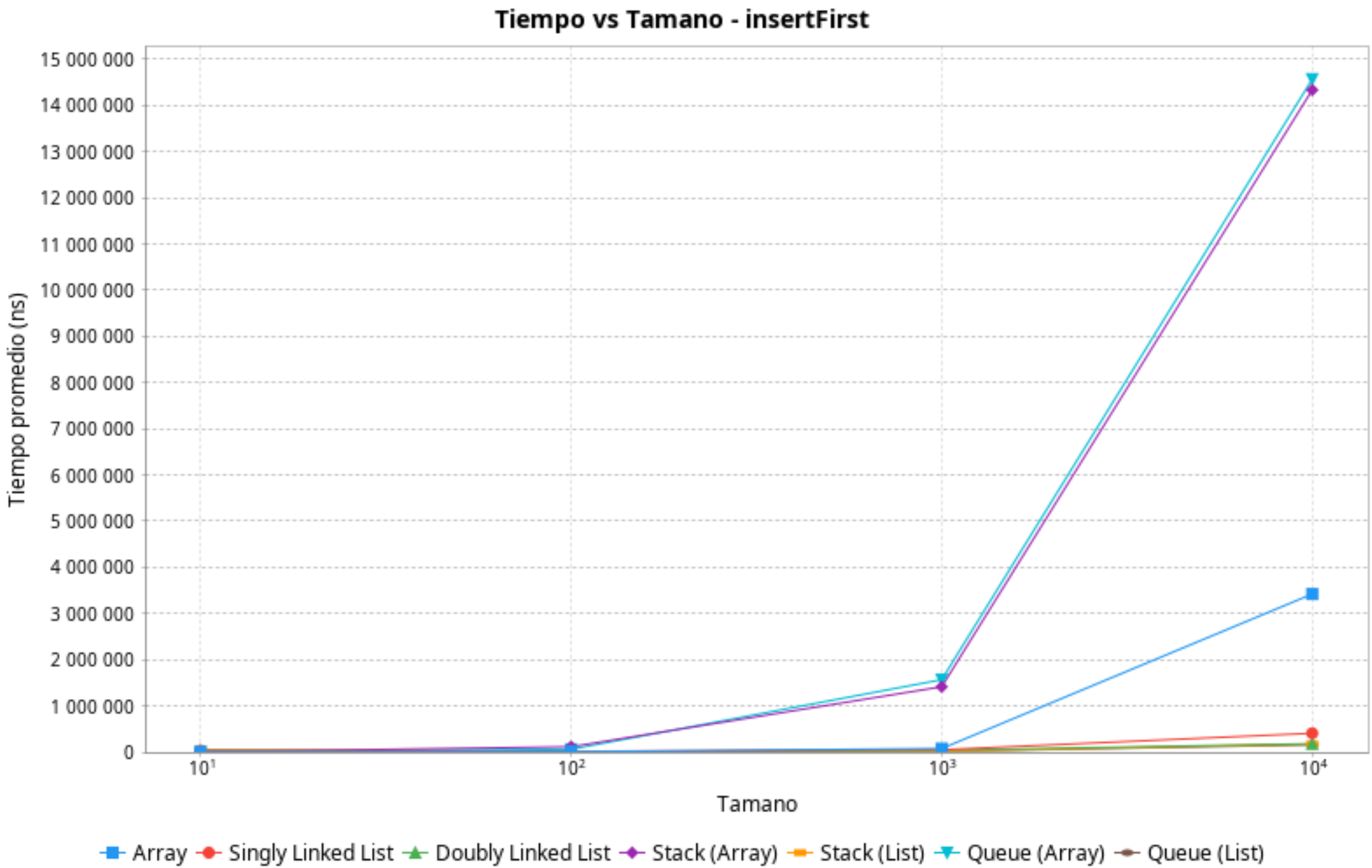
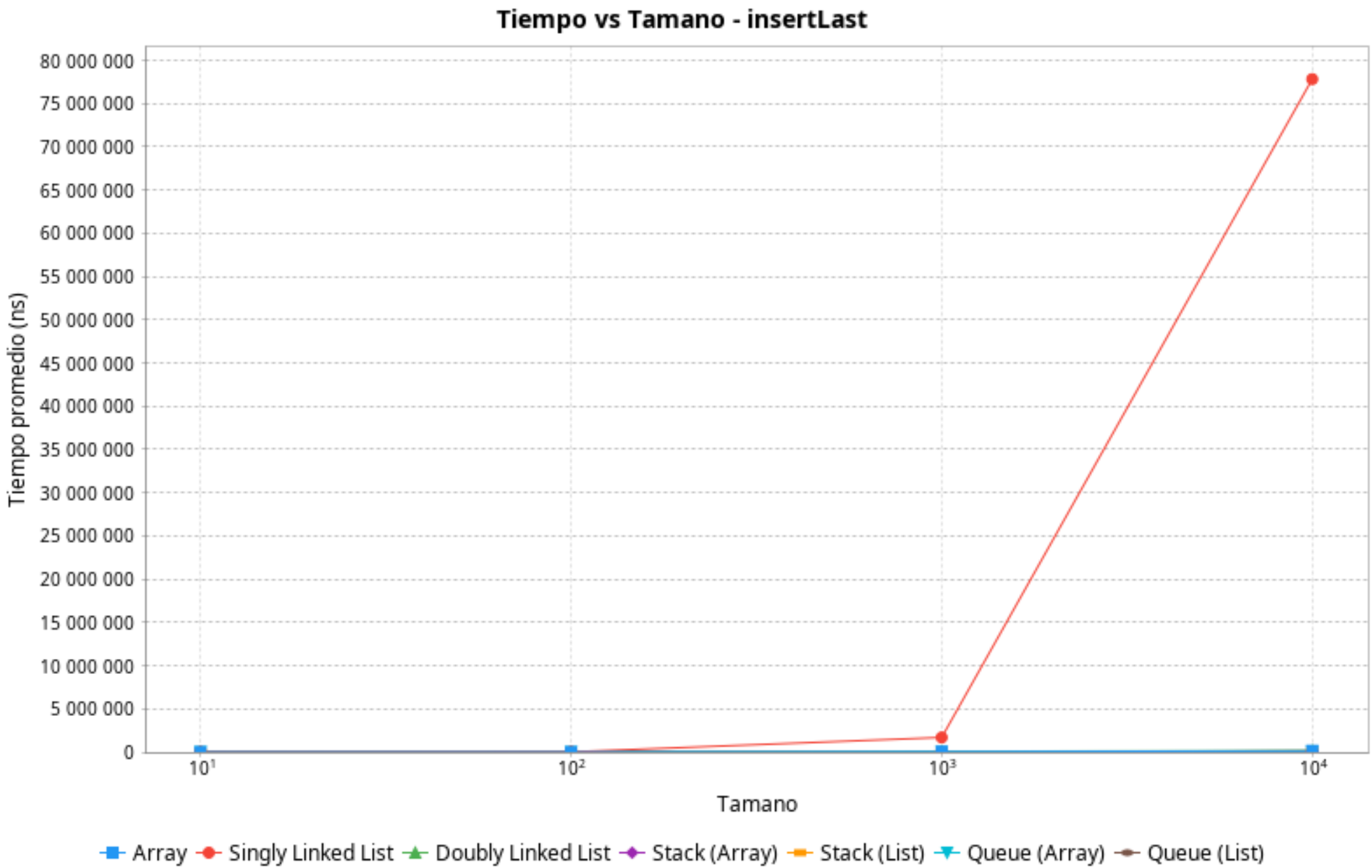
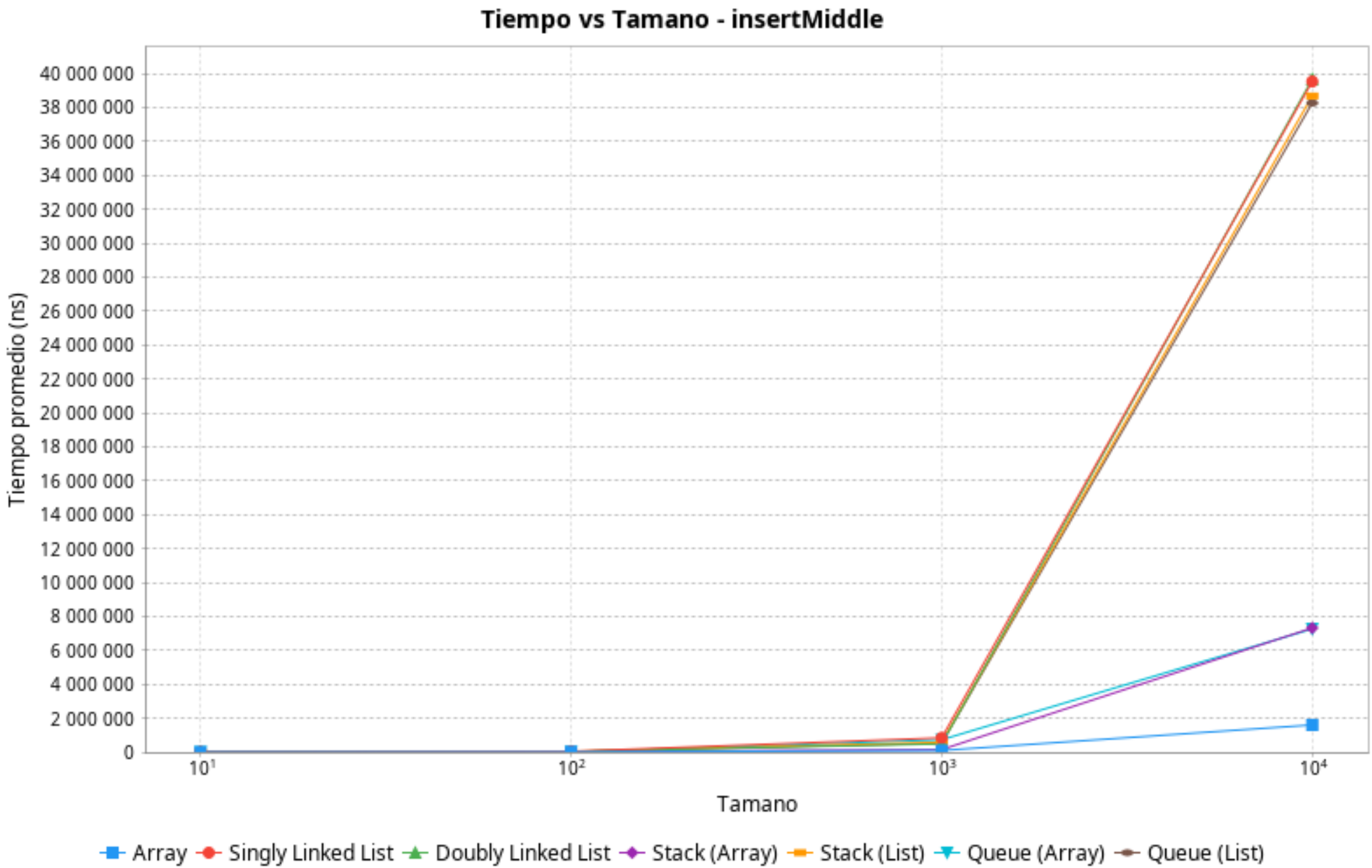
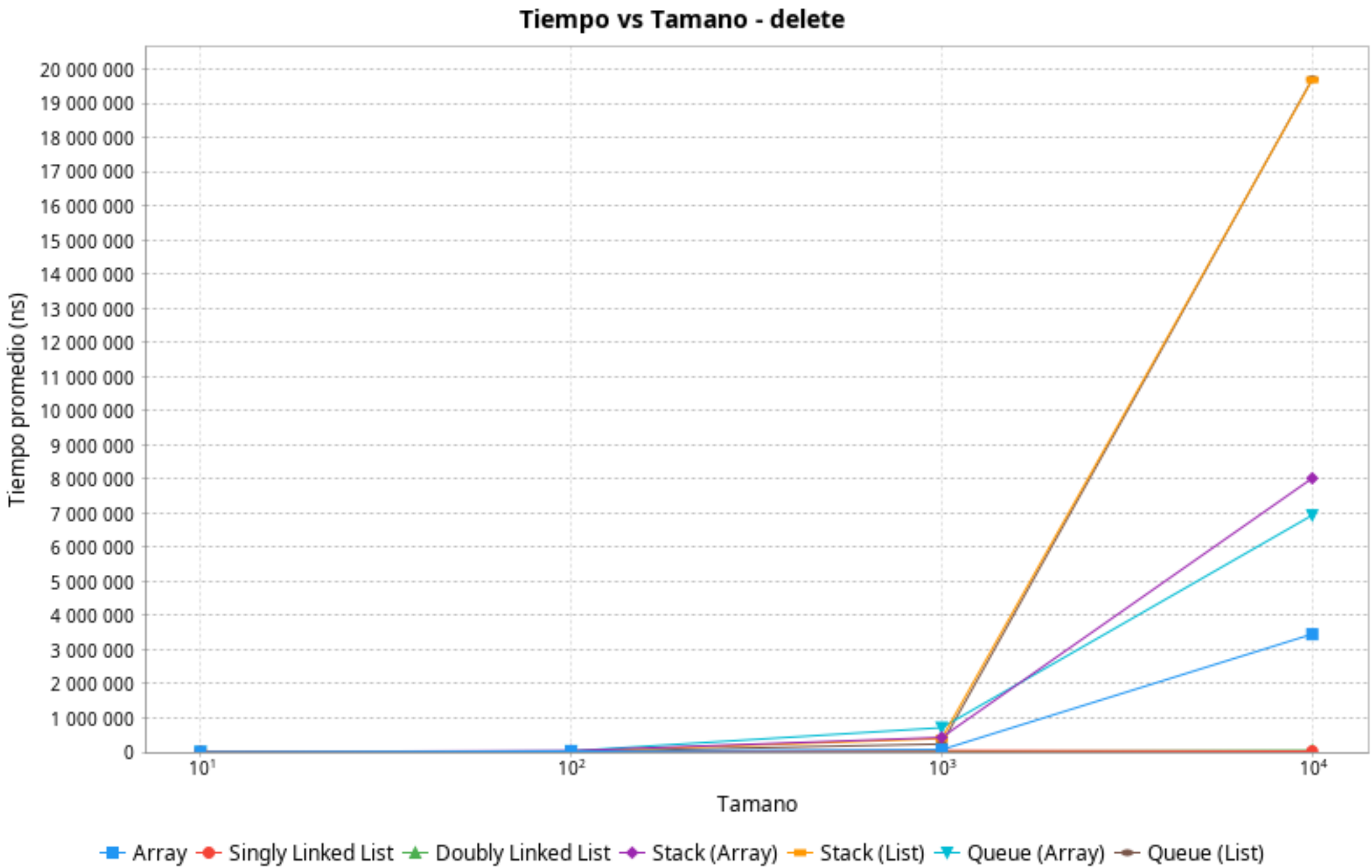
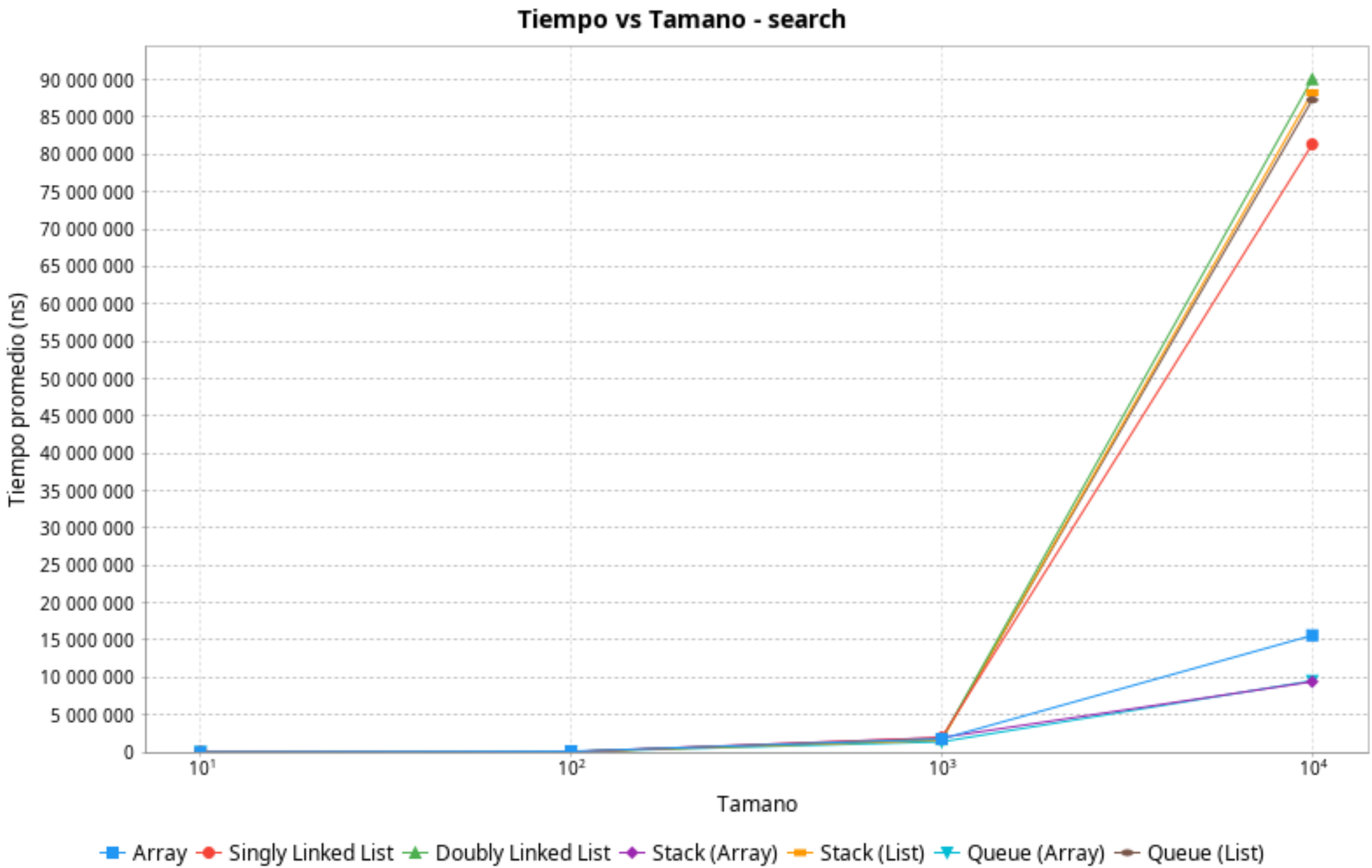
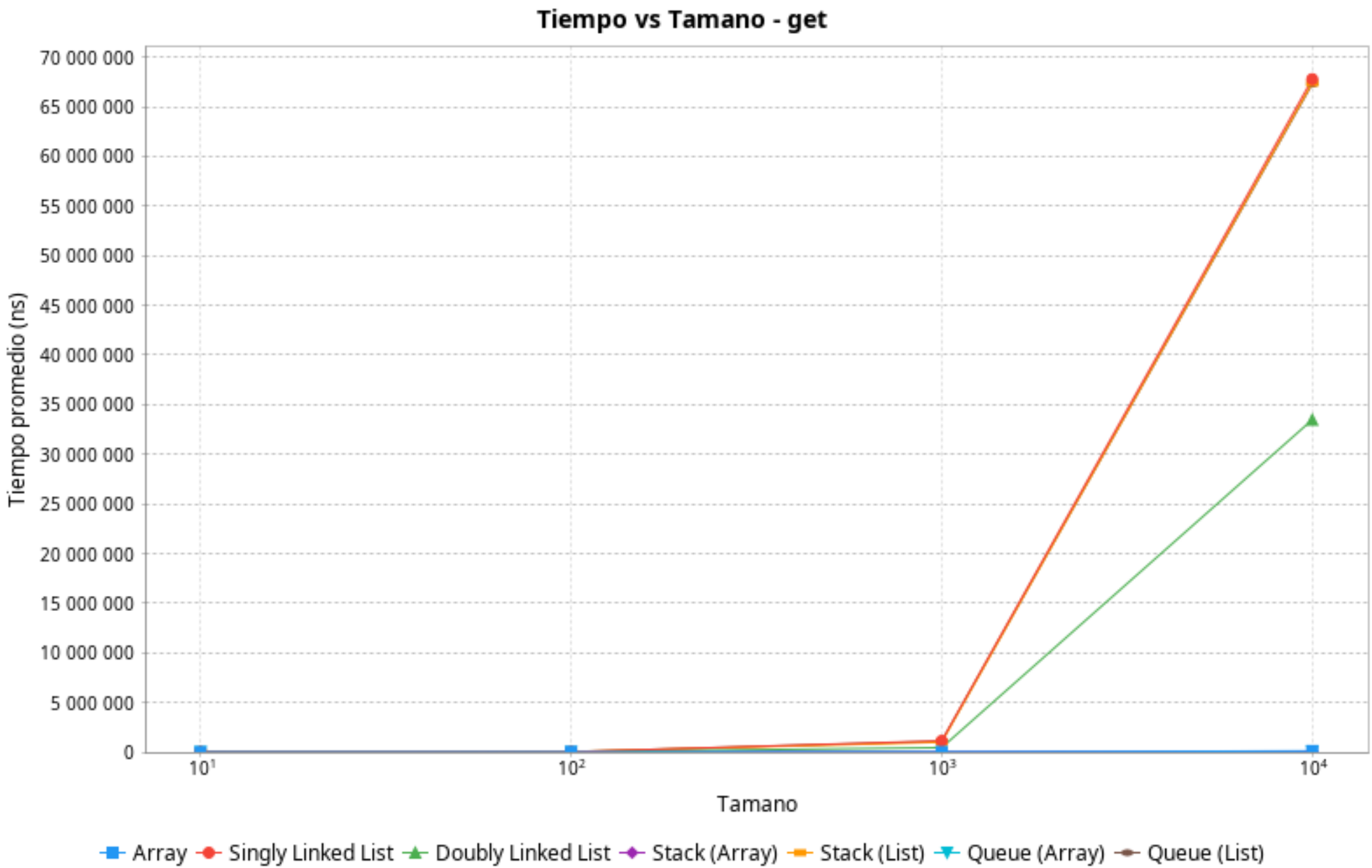

Tiempo vs Tamano - insertFirst
Tiempo promedio (ns)
15 000 000
14 000 000
13 000 000
12 000 000
11 000 000
10 000 000
9 000 000
8 000 000
7 000 000
6 000 000
5 000 000
4 000 000
3 000 000
2 000 000
1 000 000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List
Stack (Array)
Stack (List)
Queue (Array)
Queue (List)

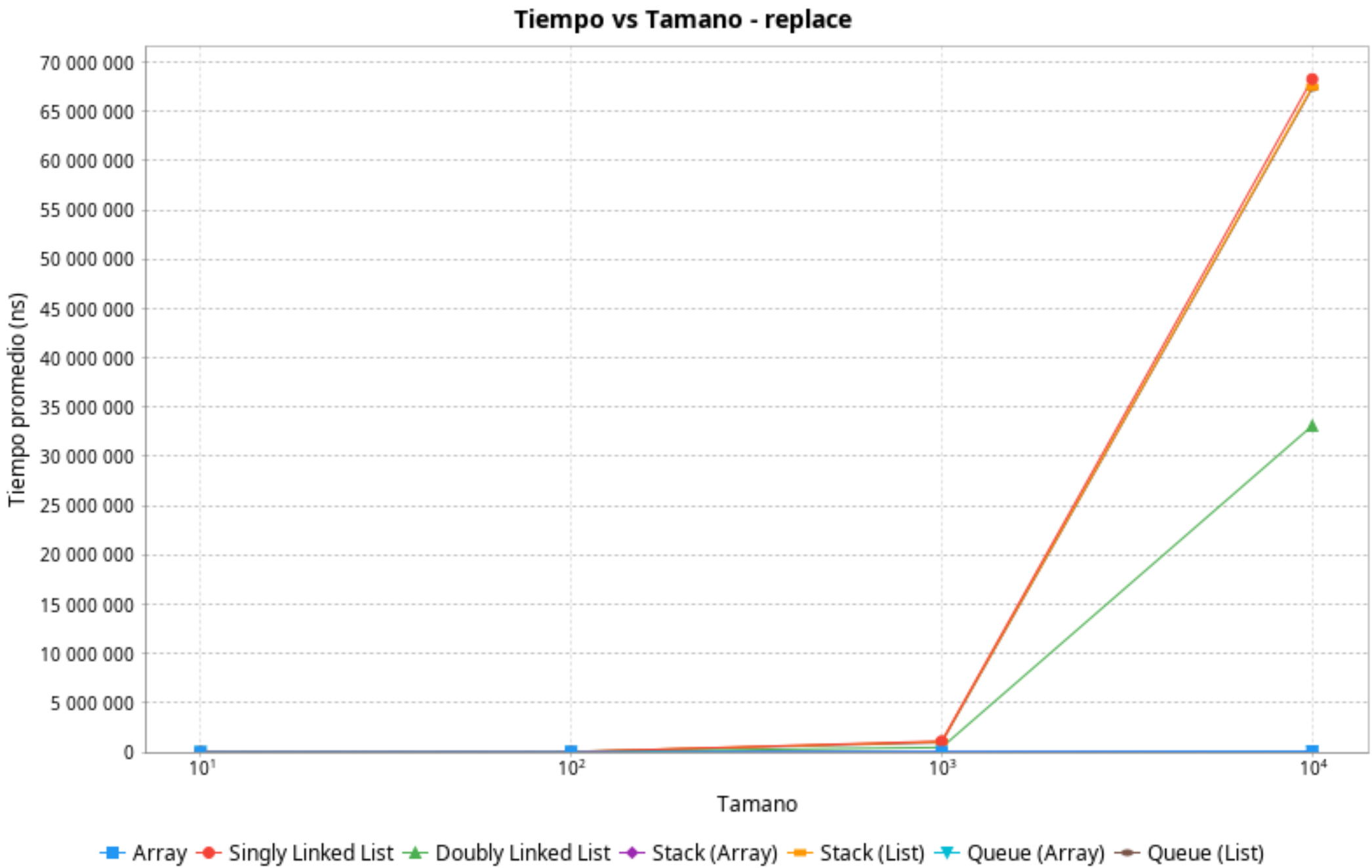

Tiempo vs Tamano - insertLast
Tiempo promedio (ns)
80 000 000
75 000 000
70 000 000
65 000 000
60 000 000
55 000 000
50 000 000
45 000 000
40 000 000
35 000 000
30 000 000
25 000 000
20 000 000
15 000 000
10 000 000
5 000 000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List
Stack (Array)
Stack (List)
Queue (Array)
Queue (List)

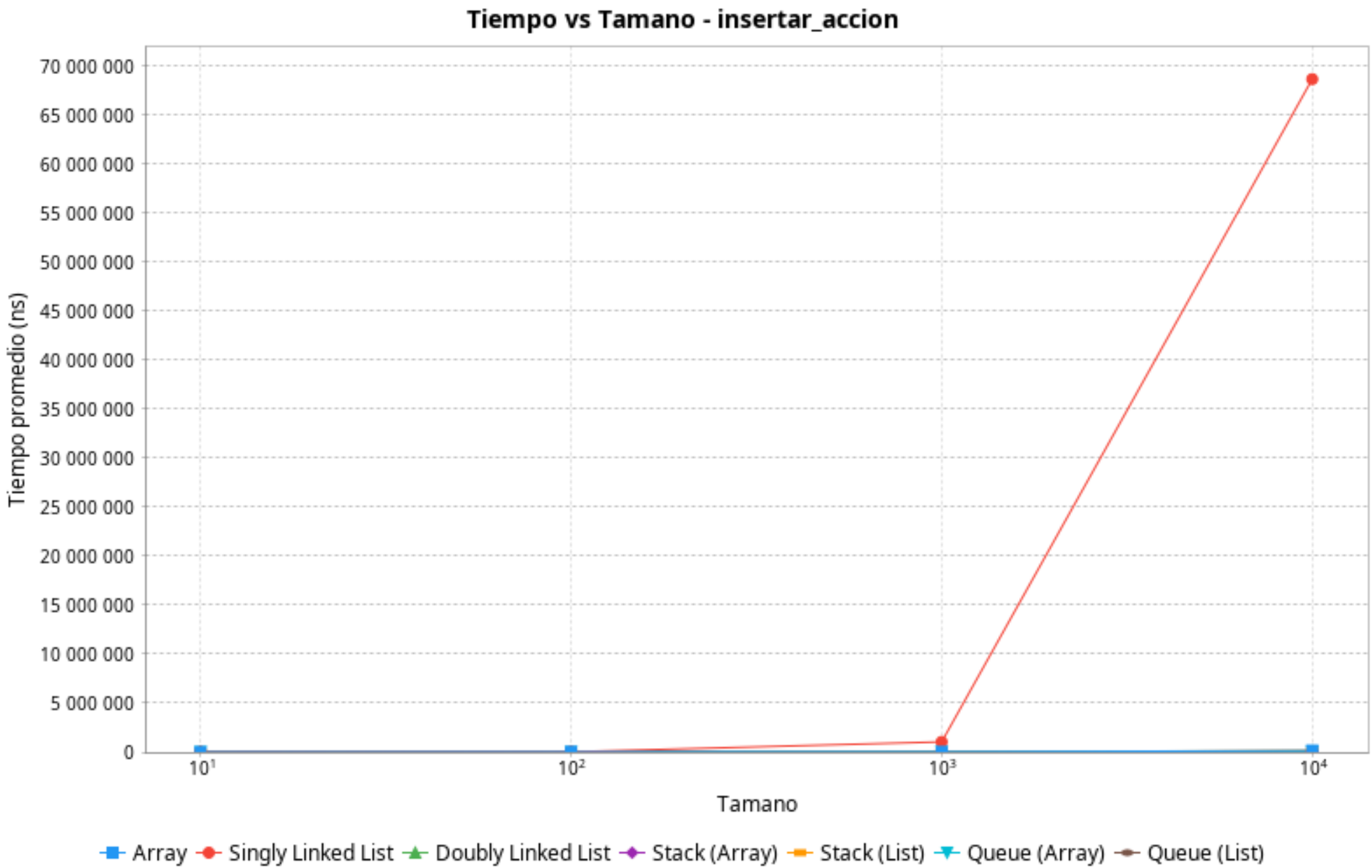

Tiempo vs Tamano - insertMiddle
Tiempo promedio (ns)
40 000 000
38 000 000
36 000 000
34 000 000
32 000 000
30 000 000
28 000 000
26 000 000
24 000 000
22 000 000
20 000 000
18 000 000
16 000 000
14 000 000
12 000 000
10 000 000
8 000 000
6 000 000
4 000 000
2 000 000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List
Stack (Array)
Stack (List)
Queue (Array)
Queue (List)

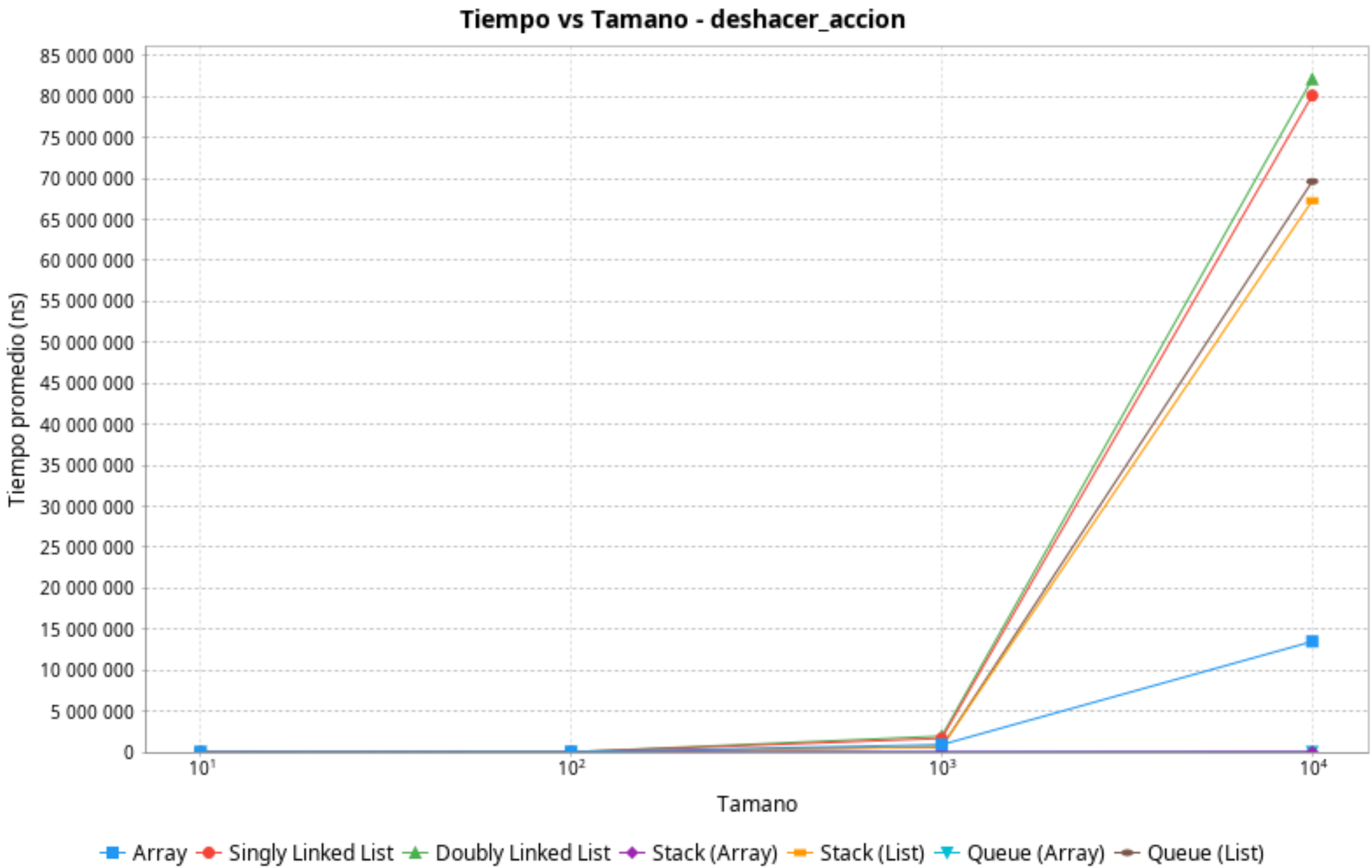

Tiempo vs Tamano - delete
Tiempo promedio (ns)
20 000 000
19 000 000
18 000 000
17 000 000
16 000 000
15 000 000
14 000 000
13 000 000
12 000 000
11 000 000
10 000 000
9 000 000
8 000 000
7 000 000
6 000 000
5 000 000
4 000 000
3 000 000
2 000 000
1 000 000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List
Stack (Array)
Stack (List)
Queue (Array)
Queue (List)

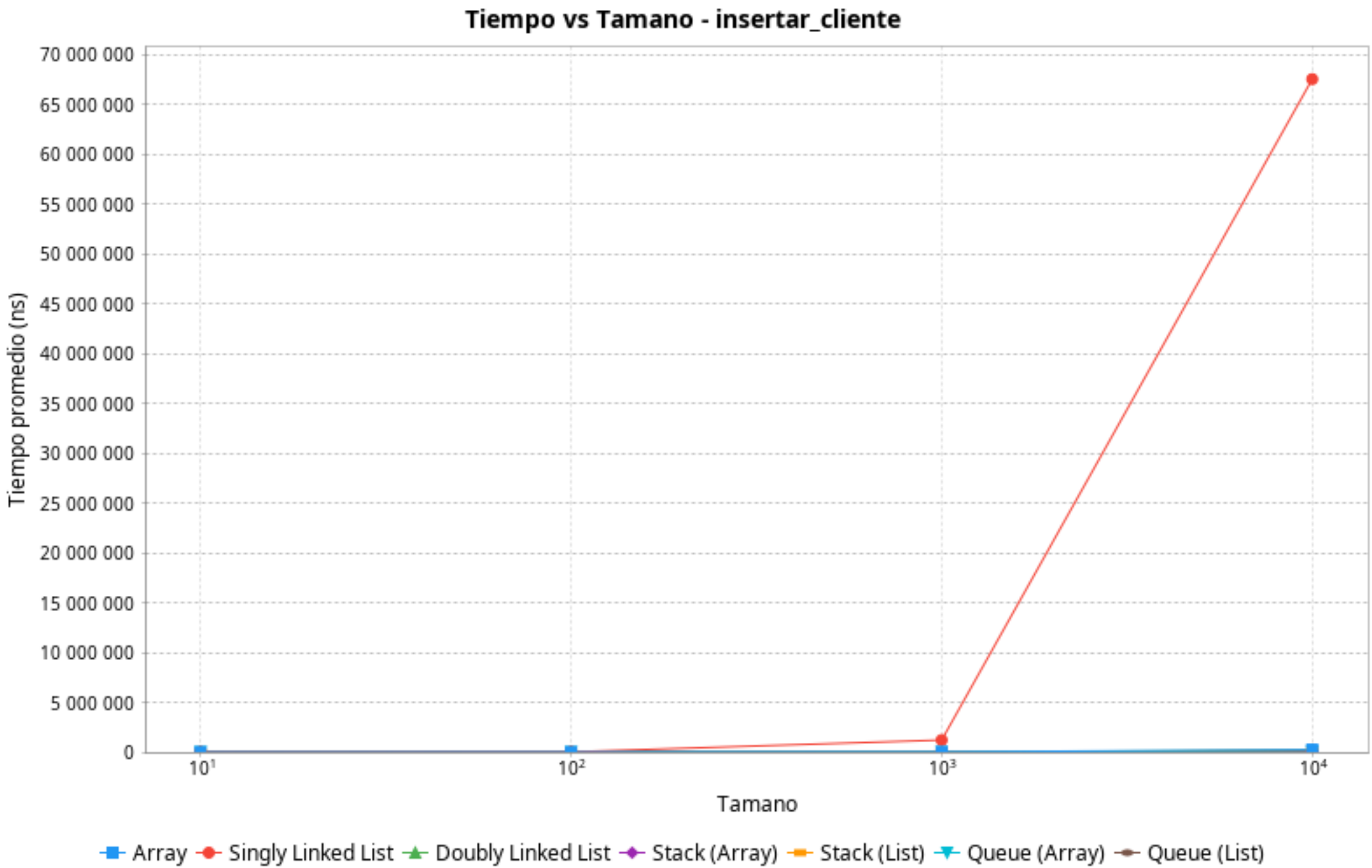

Tiempo vs Tamano - search
Tiempo promedio (ns)
90 000 000
85 000 000
80 000 000
75 000 000
70 000 000
65 000 000
60 000 000
55 000 000
50 000 000
45 000 000
40 000 000
35 000 000
30 000 000
25 000 000
20 000 000
15 000 000
10 000 000
5 000 000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List
Stack (Array)
Stack (List)
Queue (Array)
Queue (List)

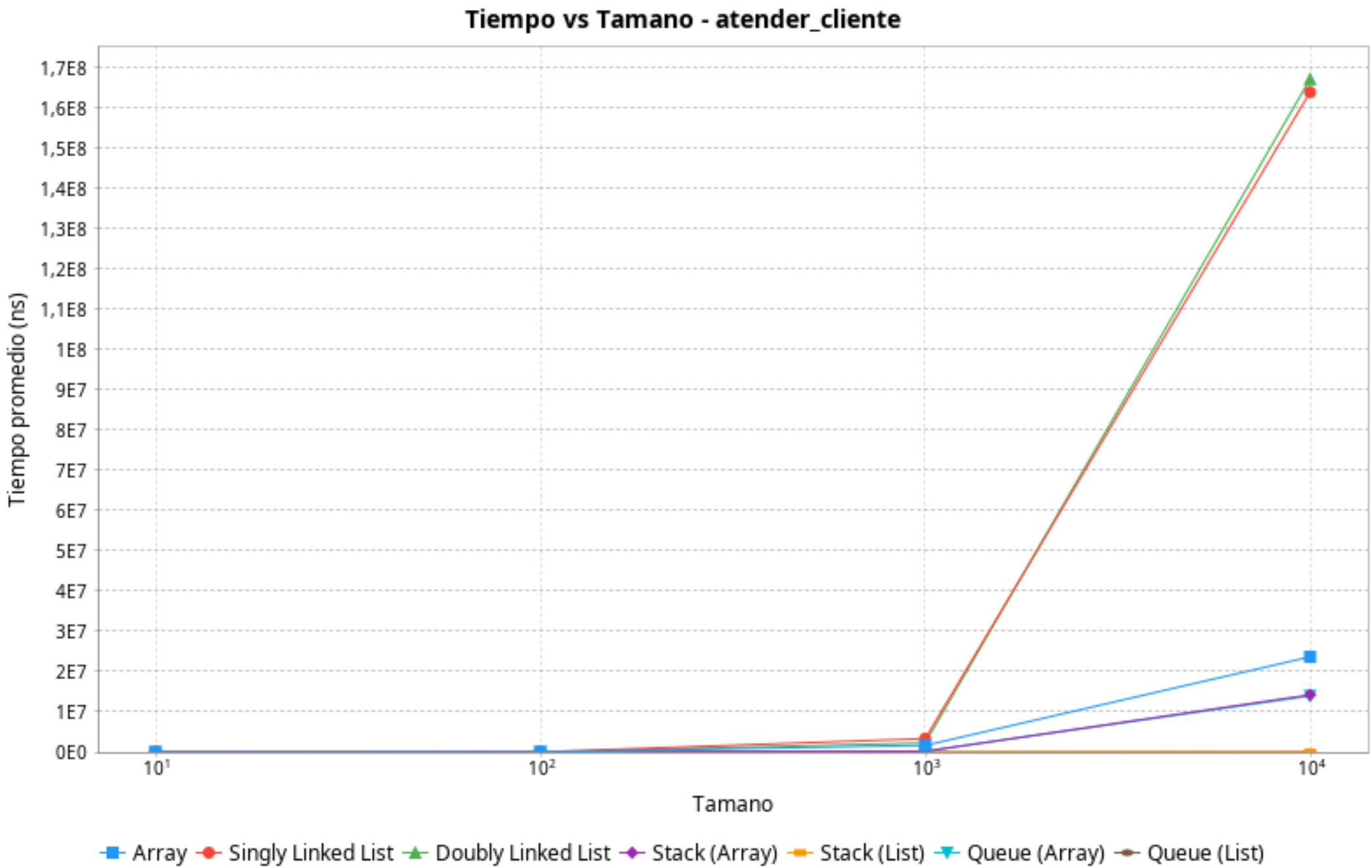

Tiempo vs Tamano - get
Tiempo promedio (ns)
70 000 000
65 000 000
60 000 000
55 000 000
50 000 000
45 000 000
40 000 000
35 000 000
30 000 000
25 000 000
20 000 000
15 000 000
10 000 000
5 000 000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List
Stack (Array)
Stack (List)
Queue (Array)
Queue (List)

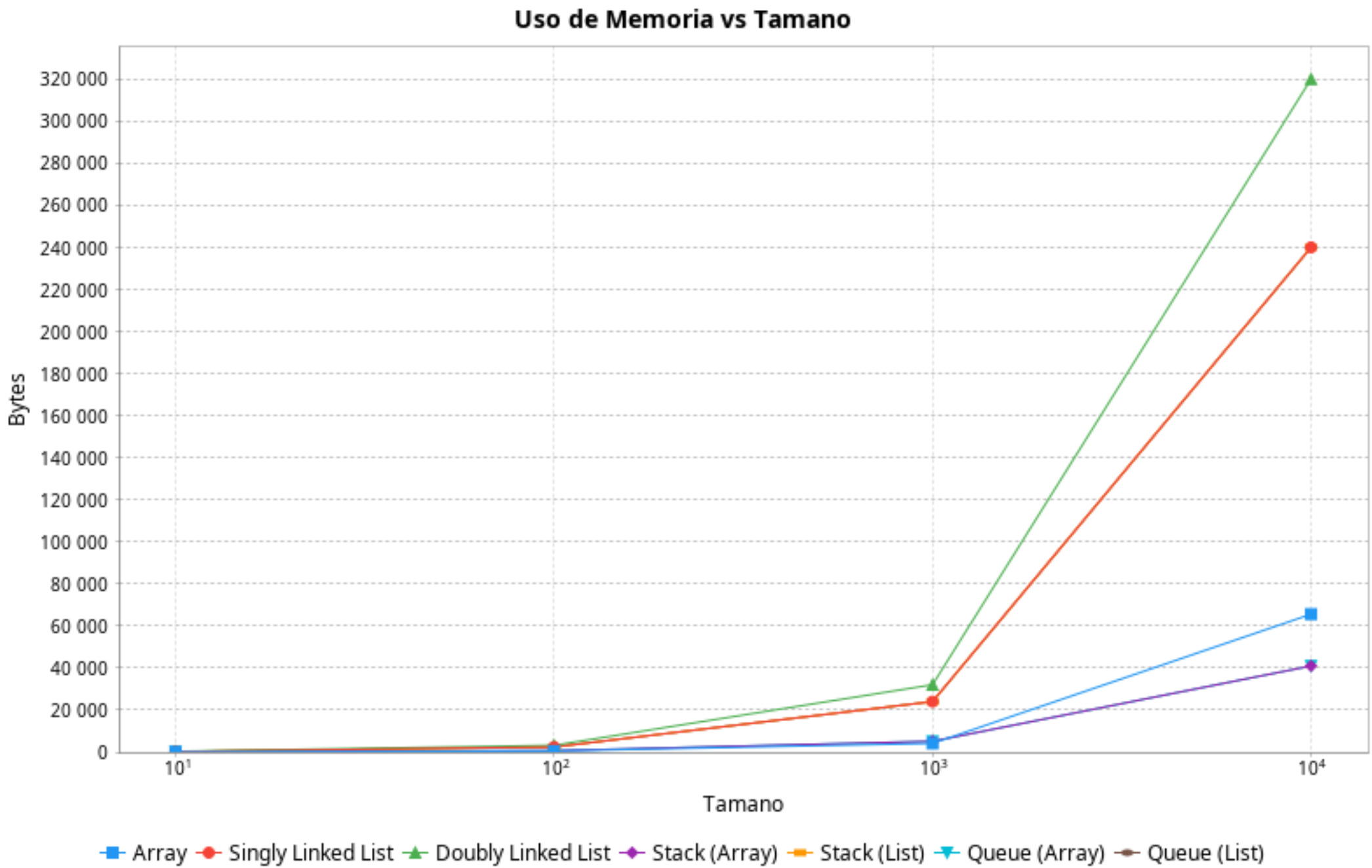

Tiempo vs Tamano - replace
Tiempo promedio (ns)
70 000 000
65 000 000
60 000 000
55 000 000
50 000 000
45 000 000
40 000 000
35 000 000
30 000 000
25 000 000
20 000 000
15 000 000
10 000 000
5 000 000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List
Stack (Array)
Stack (List)
Queue (Array)
Queue (List)

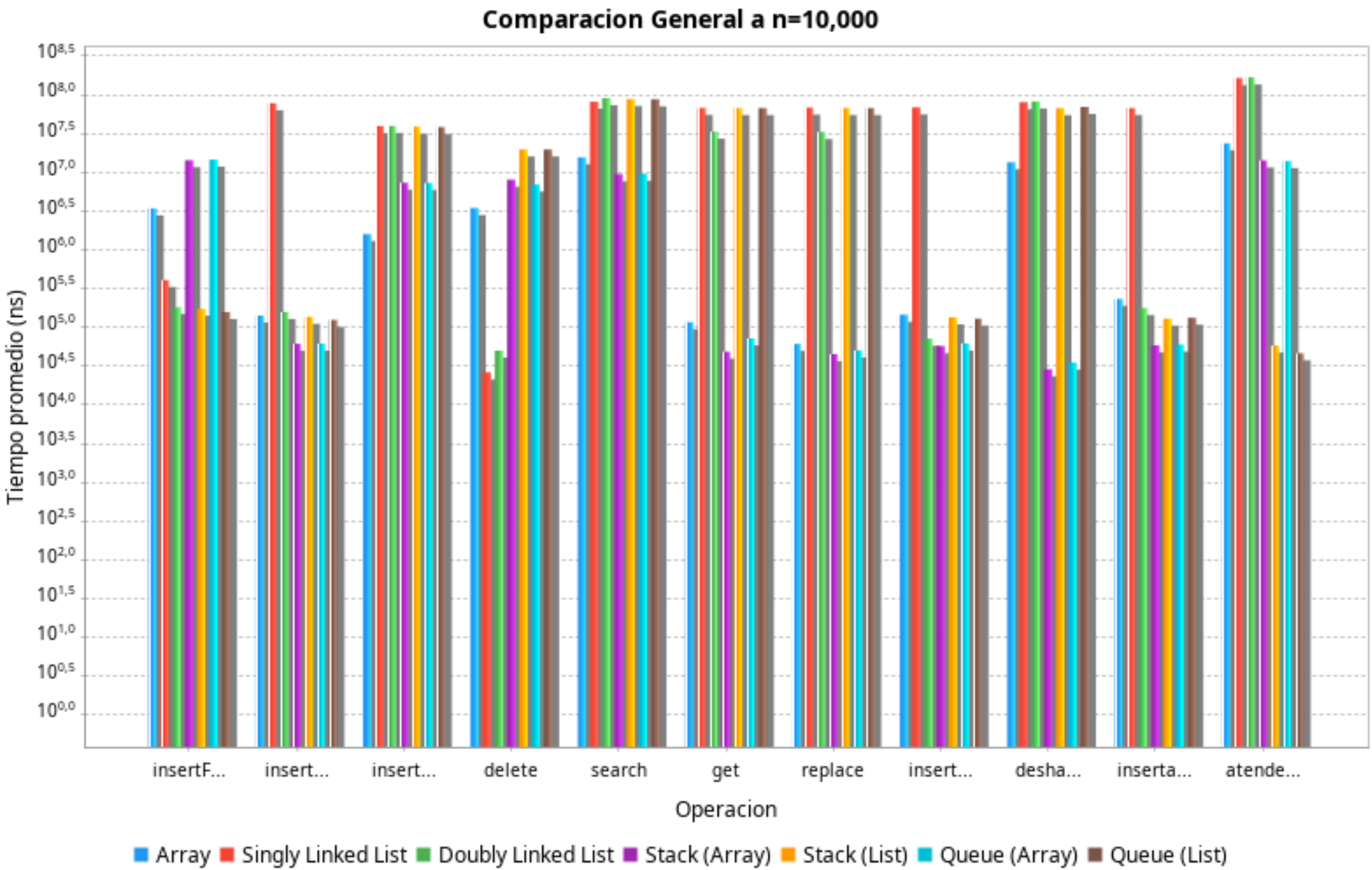

Tiempo vs Tamano - insertar_accion
Tiempo promedio (ns)
70 000 000
65 000 000
60 000 000
55 000 000
50 000 000
45 000 000
40 000 000
35 000 000
30 000 000
25 000 000
20 000 000
15 000 000
10 000 000
5 000 000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List
Stack (Array)
Stack (List)
Queue (Array)
Queue (List)


Tiempo vs Tamano - deshacer_accion
Tiempo promedio (ns)
85 000 000
80 000 000
75 000 000
70 000 000
65 000 000
60 000 000
55 000 000
50 000 000
45 000 000
40 000 000
35 000 000
30 000 000
25 000 000
20 000 000
15 000 000
10 000 000
5 000 000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List
Stack (Array)
Stack (List)
Queue (Array)
Queue (List)


Tiempo vs Tamano - insertar_cliente
Tiempo promedio (ns)
70 000 000
65 000 000
60 000 000
55 000 000
50 000 000
45 000 000
40 000 000
35 000 000
30 000 000
25 000 000
20 000 000
15 000 000
10 000 000
5 000 000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List
Stack (Array)
Stack (List)
Queue (Array)
Queue (List)


Tiempo vs Tamano - atender_cliente
Tiempo promedio (ns)
1,7E8
1,6E8
1,5E8
1,4E8
1,3E8
1,2E8
1,1E8
1E8
9E7
8E7
7E7
6E7
5E7
4E7
3E7
2E7
1E7
0E0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List
Stack (Array)
Stack (List)
Queue (Array)
Queue (List)


Uso de Memoria vs Tamano
Bytes
320 000
300 000
280 000
260 000
240 000
220 000
200 000
180 000
160 000
140 000
120 000
100 000
80 000
60 000
40 000
20 000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List
Stack (Array)
Stack (List)
Queue (Array)
Queue (List)


Comparacion General a n=10,000
Tiempo promedio (ns)
$10^{8,5}$
$10^{8,0}$
$10^{7,5}$
$10^{7,0}$
$10^{6,5}$
$10^{6,0}$
$10^{5,5}$
$10^{5,0}$
$10^{4,5}$
$10^{4,0}$
$10^{3,5}$
$10^{3,0}$
$10^{2,5}$
$10^{2,0}$
$10^{1,5}$
$10^{1,0}$
$10^{0,5}$
$10^{0,0}$
insertF...
insert...
insert...
delete
search
get
replace
insert...
desha...
inserta...
atende...
Operacion
Array
Singly Linked List
Doubly Linked List
Stack (Array)
Stack (List)
Queue (Array)
Queue (List)